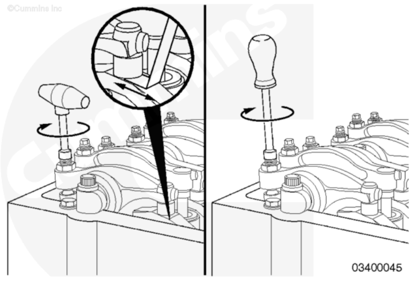
Trainings

General Information

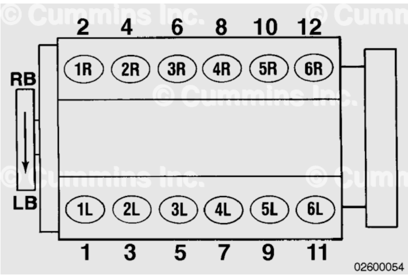
with Mechanically Actuated Injector

Cummins Inc. has found that engines in most applications will **not** experience significant valve/injector train wear. It is recommended that valve and injector adjustment is performed **only** when injectors are removed or when other repairs disturb the valve train.

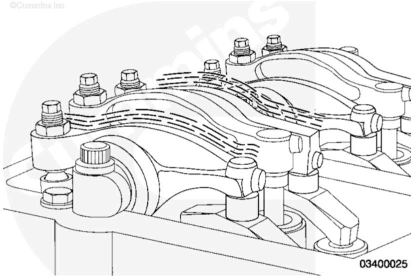


The cylinders are numbered from the front gear cover end of the engine.

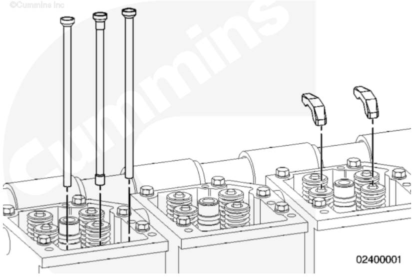
To determine the right and left banks on a QSK45 or a QSK60 engine, stand at the rear of the engine and face the front.



Each cylinder has three rocker levers. When facing the cylinder head, the rocker lever on the left is the exhaust rocker lever. The rocker lever on the right is the intake rocker lever. The center rocker lever is for the injector.



The valve and injector push rods are different. If removed, make sure the push rods are installed in the same locations from which they were removed.

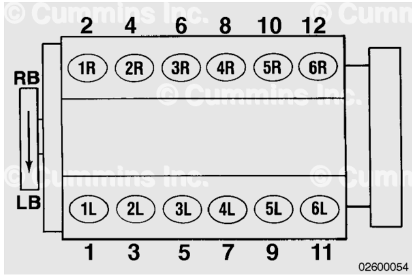


QSK45 Firing Order:

1R-6L-5R-2L-3R-4L-6R-1L-2R-5L-4R-3L

RB = Right Bank of Cylinders

LB = Left Bank of Cylinders.

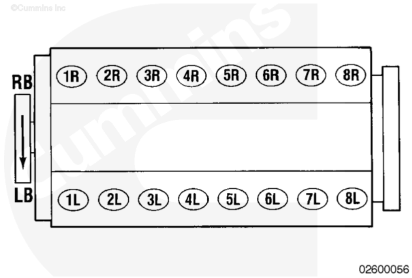


QSK60 Firing Order:

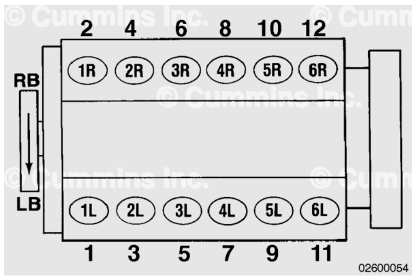
1R-1L-3R-3L-2R-2L-5R-4L-8R-8L-6R-6L-7R-7L-4R-5L

RB = Right Bank of Cylinders

LB = Left Bank of Cylinders.

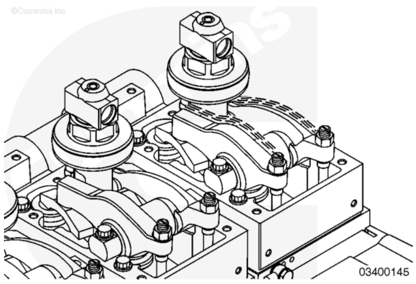


Direction of normal rotation for QSK45 and QSK60 engines is **clockwise**, when viewing from the front of the engine.



with Electronically Actuated Injector

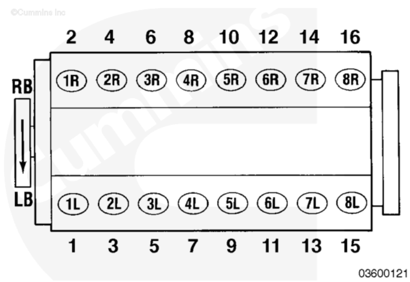
For engines with electronically actuated injectors, periodic valve adjustment is **not** required. It is recommended that the valves be adjusted **only** when an injector is removed.



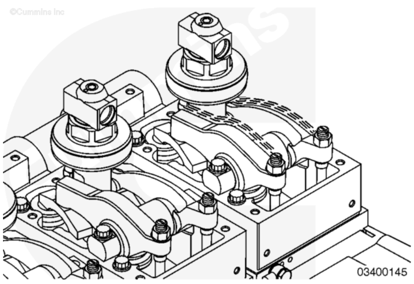
The cylinders are numbered from the front gear cover end of the engine.

To determine the right and left banks on the QSK60 engine, stand at the rear of the engine and face the front.

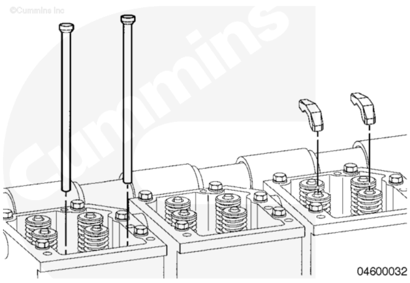
When referring to system electronics, a standard cylinder numbering has been adopted. This is **only** used for the electronic system components on engines with electronically actuated injectors.



Each cylinder has two rocker levers. When facing the cylinder head, the rocker lever on the left is the exhaust rocker lever. The rocker lever on the right is the intake rocker lever.



If the push rods were removed for service, make sure they are installed in the same locations from which they were removed.



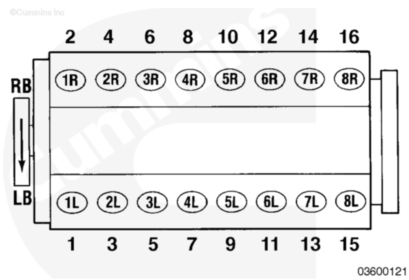
QSK60 Firing Order:

1R-1L-3R-3L-2R-2L-5R-4L-8R-8L-6R-6L-7R-7L-4R-5L or 2-1-6-5-4-3-10-7-16-15-12-11-14-13-8-9.

RB = Right Bank of cylinders

LB = Left Bank of cylinders.

Direction of normal rotation for the QSK60 engine is **clockwise**, when viewing from the front of the engine.



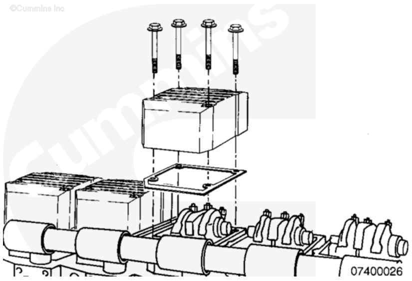
Preparatory Steps

Marine Applications

- Remove the aftercooler heat shields. Refer to Procedure 010-129 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-003-011.html)

with Mechanically Actuated Injector

- Remove the rocker lever cover and all related parts. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-003-011.html)



with Electronically Actuated Injector

- Remove the high pressure injector supply lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-051.html)
- Remove the rocker lever cover and all related parts. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-003-011.html)

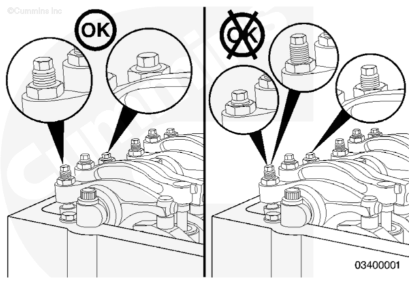
Adjust

with Mechanically Actuated Injector

Remove the rocker lever cover and all related parts. Refer to Procedure 003-011 in Section A. (/qs3/pubsys2/xml/en/procedures/56/56-003-011.html)

If the rocker lever assemblies have been removed, complete the following on all cylinders:

- Lubricate the adjusting screw threads with clean engine oil prior to making valve and injector adjustments.
- All adjusting screws **must** be loose on all cylinders, and the push rods **must** remain in alignment.
- Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods. Turn the locknuts until they touch the levers.
- The push rods will be close to the same height above the top of the rocker lever housing on the cylinder that is ready for adjustment.
- The number of threads visible above the adjusting nut will **not** be the same. There will be more threads visible on the intake adjusting screw than on the exhaust adjusting screw.

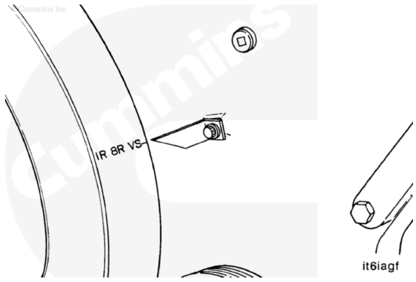


QSK45 and QSK60 engines have valve and injector adjustment marks on the vibration damper and on both sides of the flywheel housing.

Valve and injector adjustment marks **must** be aligned with the pointer, or a false adjustment can occur.

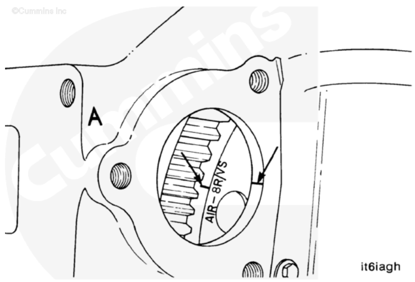
One pair of valves and one injector are adjusted at each pulley index mark, before rotating the engine to the next index mark.

Two crankshaft revolutions are required to adjust all the valves and injectors.



⚠ CAUTION ⚠

When using the starter covers on the right bank, the marks on the flywheel that begin with an A must be used or the valves and injectors will not be adjusted correctly, causing damage to the engine.



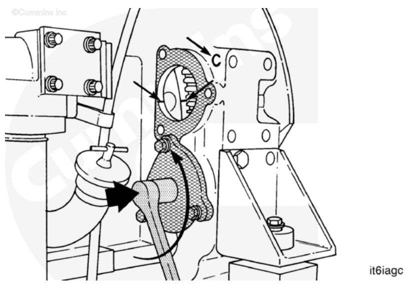
When using the valve and injector adjustment marks on the flywheel, the upper starter bore cover **must** be removed to see the marks.

The A marks **must** be used if the marks are viewed on the right banks.

The C marks **must** be used if the marks are viewed on the left bank.

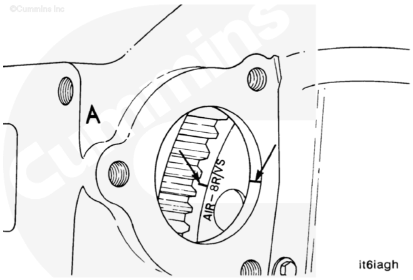
⚠ CAUTION ⚠

When using the starter covers on the left bank, the marks on the flywheel that begin with a C must be used or the valves and injectors will not be adjusted correctly, causing damage to the engine.



This illustration shows the engine barring device. To use the device, remove the clip and push the device shaft toward the flywheel. The barring device **must** be rotated **counterclockwise** to turn the flywheel and crankshaft in the direction of normal rotation.

VS represents the valve set. Ignore any top center (TC) marks while setting the valves and injectors.

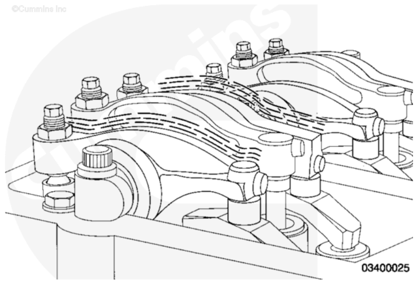


Determine the cylinder in position for valve set:

The valves are ready to be adjusted on the cylinder that has all of the valves closed.

Note : When all of the valves are closed, there is some movement in the rocker levers.

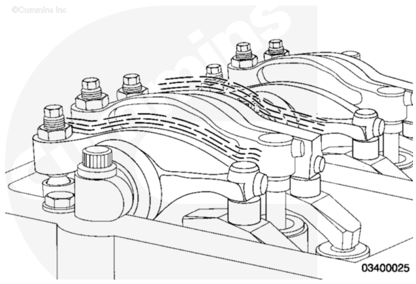
Check the two cylinders shown on the VS mark.



If the rocker levers have **not** been removed, wiggle the valve rocker levers on the two cylinders in question. Set the valves on the cylinder where both levers feel loose.

Adjustment can begin on any valve set mark.

Use the following chart to determine the injector that is ready to adjust.



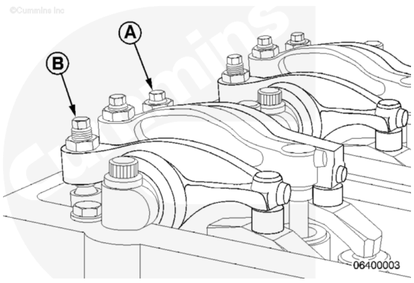
QSK45 Outer Base Circle Valve and Injector Set			
VS Mark	**Valves Closed on Cylinder Number**	**Adjust Valves on Cylinder Number**	**Adjust Injector on Cylinder Number**
1R-6R VS	1R	1R	2R
6L-1L VS	6L	6L	5L
5R-2R VS	5R	5R	4R
2L-5L VS	2L	2L	3L
3R-4R VS	3R	3R	1R
4L-3L VS	4L	4L	6L
1R-6R VS	6R	6R	5R
6L-1L VS	1L	1L	2L
5R-2R VS	2R	2R	3R
2L-5L VS	5L	5L	4L
3R-4R VS	4R	4R	6R
4L-3L VS	3L	3L	1L

QSK60 Outer Base Circle Valve and Injector Set			
VS Mark	**Valves Closed on Cylinder Number**	**Adjust Valves on Cylinder Number**	**Adjust Injector on Cylinder Number**
1R-8R VS	1R	1R	6R
1L-8L VS	1L	1L	6L
3R-6R VS	3R	3R	7R
3L-6L VS	3L	3L	7L
2R-7R VS	2R	2R	4R
2L-7L VS	2L	2L	5L
4R-5R VS	5R	5R	1R
4L-5L VS	4L	4L	1L
1R-8R VS	8R	8R	3R
1L-8L VS	8L	8L	3L
3R-6R VS	6R	6R	2R
3L-6L VS	6L	6L	2L
2R-7R VS	7R	7R	5R
2L-7L VS	7L	7L	4L
4R-5R VS	4R	4R	8R
4L-5L VS	5L	5L	8L

Valves

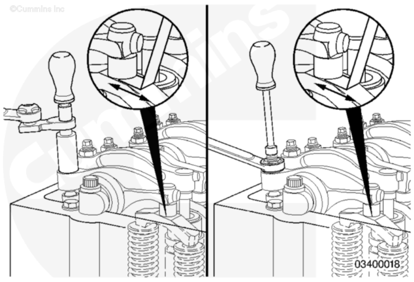
Valve Clearances - Initial Set			
	mm		**in**
Exhaust Valves (A)	0.81	max	0.032
Intake Valves (B)	0.36	max	0.014

Valve Clearances - Recheck			
	mm		**in**
Exhaust Valves (A)	0.74	min	0.029
	0.89	max	0.035
Intake Valves (B)	0.36	min	0.014
	0.51	max	0.020



There are two different methods used to set valve lash clearance: Torque wrench method and feeler gauge method; both are described below. Either method can be used. However, the torque wrench method has proven to be the most consistent.

Make sure the crossheads are firmly in place on the valve stems.

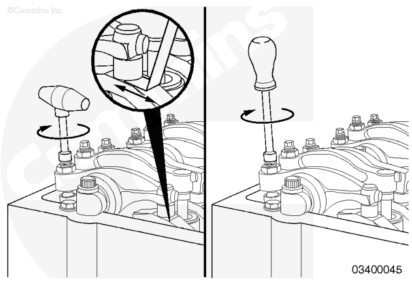


Select the proper feeler gauge for the valves being set. Use service tool, Part Number 3824901, or equivalent.

Make sure the feeler gauge is under the center of the ball-and-socket, or the socket can rock or tip, resulting in an incorrect adjustment. Hold the swivel foot flat when checking the lash, to avoid false readings.

The adjustment screws **must** turn freely, or a false reading or setting can occur.

Valve Adjustment - Torque Wrench Method:



Make sure the parts are aligned, and squeeze the oil out of the valve and the injector train by tightening the adjusting screw.

Loosen the adjusting screw at least one revolution.

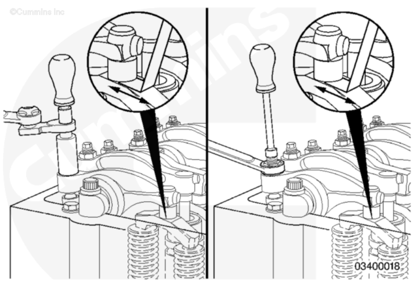
Insert the feeler gauge between the rocker lever socket and the crosshead.

Use torque wrench, Part Number 3376592, and tighten the adjusting screw.

Torque Value: 0.7 n•m [6 in-lb]

Remove the feeler gauge.

The adjusting screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without a torque wrench adapter, Part Number 3163196.



Tighten the locknut.

Torque Value:
With Adapter 84 n•m [62 ft-lb]

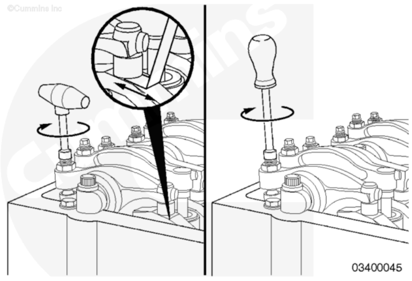
Torque Value:
Without Adapter 105 n•m [77 ft-lb]

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct if the thicker feeler gauge will fit.

Repeat the adjustment process until the proper lash is obtained.

Valve Adjustment - Feeler Gauge Method:

Make sure parts are aligned, and squeeze the oil out of the valve and injector train by tightening the adjusting screw.

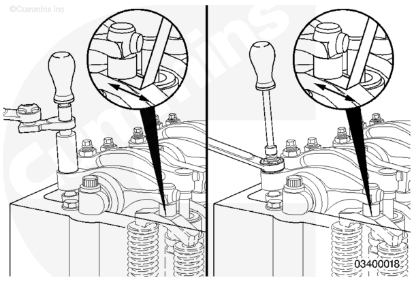


Loosen the adjusting screw at least one revolution.

Insert the feeler gauge between the rocker lever socket and the crosshead.

Tighten the adjusting screw until the rocker lever touches the feeler gauge.

The adjusting screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without a torque wrench adapter, Part Number 3163196.



Tighten the locknut.

Torque Value:
With Adapter 84 n•m [62 ft-lb]

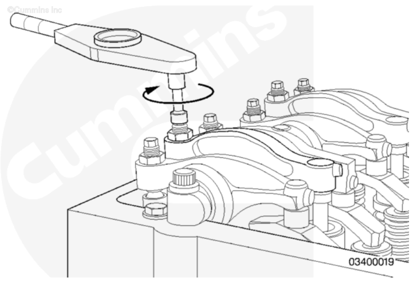
Torque Value:
Without Adapter 105 n•m [77 ft-lb]

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct if the thicker feeler gauge will fit.

Repeat the adjustment process until the proper lash is obtained.

Injectors

Do **not** use a click-type torque wrench.



Use a dial-type torque wrench, Part Number 3164795, with a range up to 34 N•m [25 ft-lb] to tighten the injector rocker lever adjusting screw. If the screw chatters during setting, repair the screw and lever as required.

Hold the torque wrench in a position that allows you to look in a direct line at the dial. This is to make sure the dial will be read accurately.

Make sure the parts are aligned, and squeeze the oil out of the valve and injector train by tightening the adjusting screw.

Use this initial adjustment to preload the valve train and injector.

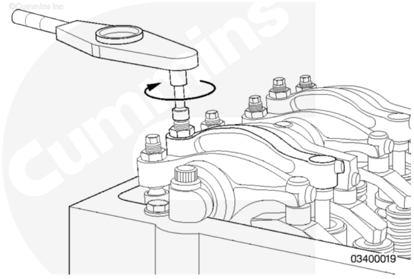
Tighten the injector adjusting screw.

Torque Value: 28 n•m [248 in-lb]

Loosen the adjusting screw at least one revolution.

Tighten the adjusting screw.

Torque Value: 19 n•m [168 in-lb]

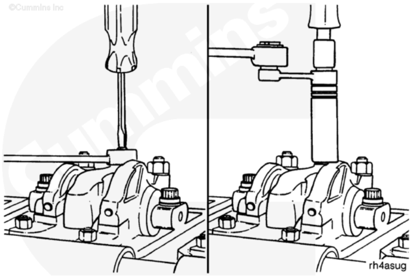


The adjusting screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without a torque wrench adapter, Part Number 3163196.

Tighten the locknut.

Torque Value:
With Adapter 84 n•m [62 ft-lb]

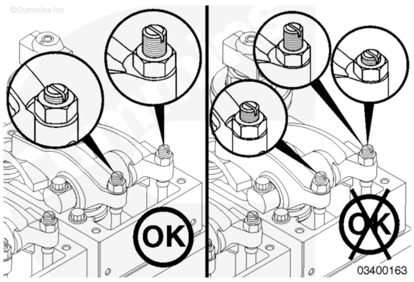
Torque Value:
Without Adapter 105 n•m [77 ft-lb]



with Electronically Actuated Injector

If the rocker lever assemblies have been removed, complete the following on both cylinders:

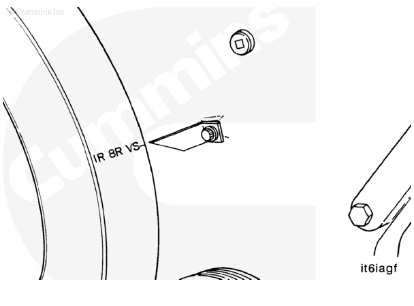
- Lubricate the adjusting screw threads with clean engine oil prior to making valve and injector adjustments.
- All adjusting screws **must** be loose on all cylinders, and the push rods **must** remain in alignment.
- Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods. Turn the locknuts until they touch the levers.
- The push rods will be close to the same height above the top of the rocker lever housing on the cylinder that is ready for adjustment.
- The number of threads visible above the adjusting nut will **not** be the same. There will be more threads visible on the intake adjusting screw than on the exhaust adjusting screw.



QSK60 engines have valve adjustment marks on the vibration damper and on both sides of the flywheel housing.

Valve adjustment marks **must** be aligned with the pointer or a false adjustment can occur.

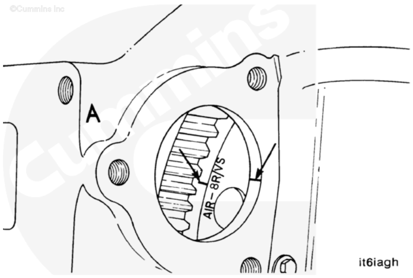
One pair of valves are adjusted at each pulley index mark, before rotating the engine to the next index mark.



Two crankshaft revolutions are required to adjust all of the valves.

⚠ CAUTION ⚠

When using the starter covers on the right bank, the marks on the flywheel that begin with an A must be used or the valves will not be adjusted correctly, causing damage to the engine.



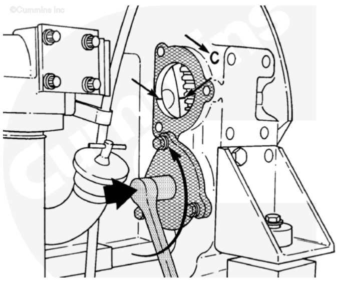
When using the valve adjustment marks on the flywheel, the upper starter bore cover **must** be removed to see the marks.

The A marks **must** be used if the marks are viewed on the right bank.

The C marks **must** be used if the marks are viewed on the left bank.

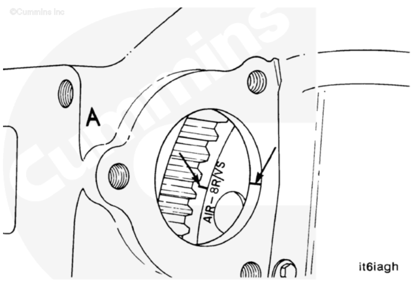
⚠ CAUTION ⚠

When using the starter covers on the left bank, the marks on the flywheel that begin with a C must be used or the valves will not be adjusted correctly, causing damage to the engine.



This illustration shows the engine barring device. To use the device, remove the clip and push the device shaft toward the flywheel. The barring device **must** be rotated **counterclockwise** to turn the flywheel and crankshaft in the direction of normal rotation.

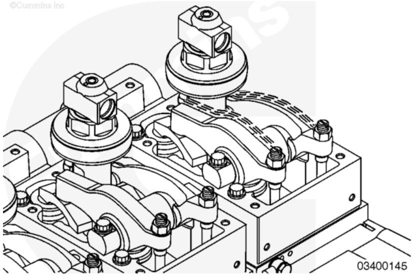
VS represents the valve set. Ignore any TC (top center) marks while setting the valves.



Determine the cylinder in position for valve set:

The valves are ready to be adjusted on the cylinder that has all of the valves closed.

Note : When all of the valves are closed, there is some movement in the rocker levers.



Check the two cylinders shown on the VS mark.

If the rocker levers have **not** been removed, wiggle the valve rocker levers on the two cylinders in question. Set the valves on the cylinder where both levers feel loose.

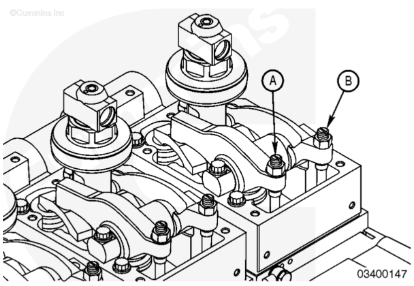
Adjustment can begin on any valve set mark.

Use the following chart to determine the cylinder that is ready to adjust.

QSK60 Outer Base Circle Valve Set		
VS Mark	**Valves Closed on Cylinder Number**	**Adjust Valves on Cylinder Number**
1R-8R VS	1R	1R
1L-8L VS	1L	1L
3R-6R VS	3R	3R
3L-6L VS	3L	3L
2R-7R VS	2R	2R
2L-7L VS	2L	2L
4R-5R VS	5R	5R
4L-5L VS	4L	4L
1R-8R VS	8R	8R
1L-8L VS	8L	8L
3R-6R VS	6R	6R
3L-6L VS	6L	6L
2R-7R VS	7R	7R
2L-7L VS	7L	7L
4R-5R VS	4R	4R
4L-5L VS	5L	5L

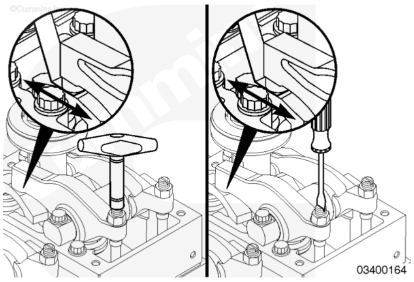
Valves

Valve Clearances - Initial Set			
	mm		**in**
Exhaust Valves (A)	0.81	MAX	0.032
Intake Valves (B)	0.36	MAX	0.014



Valve Clearances - Recheck			
	mm		in
Exhaust Valves (A)	0.74	min	0.029
	0.89	max	0.035
Intake Valves (B)	0.36	min	0.014
	0.51	max	0.020

There are two different methods used to set valve lash clearance: Torque wrench method and feeler gauge method; both are described below. Either method can be used. However, the torque wrench method has proven to be the most consistent.

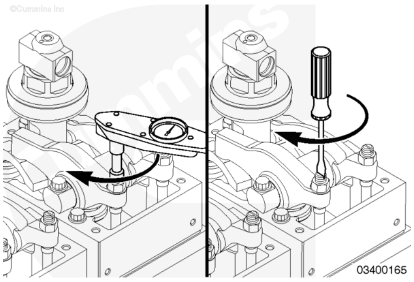


Make sure the crossheads are firmly in place on the valve stems.

Select the proper feeler gauge for the valves being set. Use service tool, Part Number 3824901, or equivalent.

Make sure the feeler gauge is under the center of the ball-and-socket, or the socket can rock or tip, resulting in an incorrect adjustment. Hold the swivel foot flat when checking the lash to avoid false readings.

The adjustment screws **must** turn freely, or a false reading or setting can occur.



Valve Adjustment - Torque Wrench Method:

Make sure the parts are aligned, and squeeze the oil out of the valve train by tightening the adjusting screw.

Loosen the adjusting screw at least one revolution.

Insert the feeler gauge between the rocker lever socket and the crosshead.

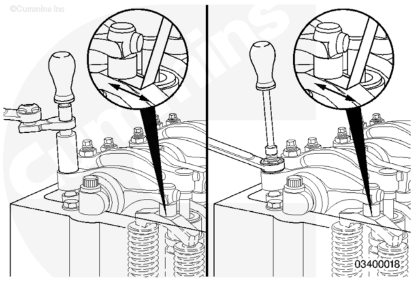
Use torque wrench, Part Number 3376592, and tighten the adjusting screw.

Torque Value: 0.7 n•m [6 in-lb]

Remove the feeler gauge.

The adjusting screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without torque wrench adapter, Part Number 3163196.

Tighten the locknut.



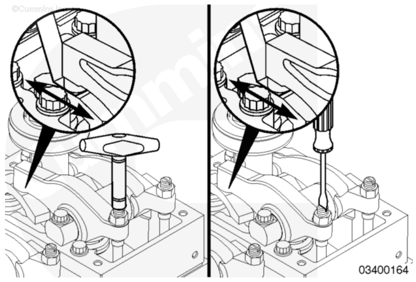
Torque Value:
With Adapter 84 n•m [62 ft-lb]

Torque Value:
Without Adapter 105 n•m [77 ft-lb]

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct if the thicker feeler gauge will fit.

Repeat the adjustment process until the proper lash is obtained.

Valve Adjustment - Feeler Gauge Method:



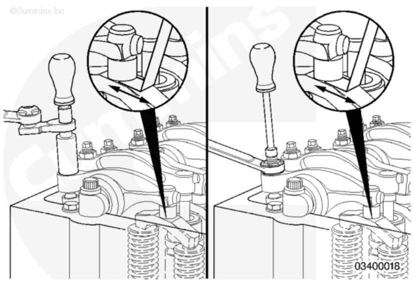
Make sure parts are aligned, and squeeze the oil out of the valve and injector train by tightening the adjusting screw.

Loosen the adjusting screw at least one revolution.

Insert the feeler gauge between the rocker lever socket and the crosshead.

Tighten the adjusting screw until the rocker lever touches the feeler gauge.

The adjusting screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without a torque wrench adapter, Part Number 3163196.



Tighten the locknut.

Torque Value:
With Adapter 84 n•m [62 ft-lb]

Torque Value:
Without Adapter 105 n•m [77 ft-lb]

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct if the thicker feeler gauge will fit.

Repeat the adjustment process until the proper lash is obtained.

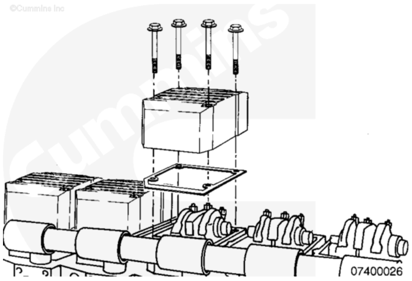
Finishing Steps

Marine Applications

- Install the rocker lever cover and all related parts. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-003-011.html)
- Install the aftercooler heat shields. Refer to Procedure 010-129 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html)
- Operate the engine. Check for leaks.

with Mechanically Actuated Injector

- Install the rocker lever cover and all related parts. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-003-011.html)
- Operate the engine. Check for leaks.



with Electronically Actuated Injector

- Install the rocker lever cover and all related parts. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/56/56-003-011.html)
- Install the high pressure injector supply lines. Refer to Procedure 006-051 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-051.html)
- Operate the engine. Check for leaks.

Last Modified: 11-Oct-2022